📌 STEP 1: Import Libraries

In [56]:
```
# Importing necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
from sklearn.pipeline import Pipeline
```

STEP 2: Load Dataset

In [14]:
```
# Step 2: Load the data file using pandas
df = pd.read_csv("googleplaystore.csv")

# Checking the first few rows
df.head()
```

Out[14]:

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Current Ver	Android Ver
0	Photo Editor & Candy Camera & Grid & ScrapBook	ART_AND_DESIGN	4.1	159	19M	10,000+	Free	0	Everyone	Art & Design	January 7, 2018	1.0.0	4.0.3 and up
1	Coloring book moana	ART_AND_DESIGN	3.9	967	14M	500,000+	Free	0	Everyone	Art & Design;Pretend Play	January 15, 2018	2.0.0	4.0.3 and up
2	U Launcher Lite – FREE Live Cool Themes, Hide ...	ART_AND_DESIGN	4.7	87510	8.7M	5,000,000+	Free	0	Everyone	Art & Design	August 1, 2018	1.2.4	4.0.3 and up
3	Sketch - Draw & Paint	ART_AND_DESIGN	4.5	215644	25M	50,000,000+	Free	0	Teen	Art & Design	June 8, 2018	Varies with device	4.2 and up
4	Pixel Draw - Number Art Coloring Book	ART_AND_DESIGN	4.3	967	2.8M	100,000+	Free	0	Everyone	Art & Design;Creativity	June 20, 2018	1.1	4.4 and up

📌 STEP 3: Check Null Values

In [15]:

```
# Step 3: Check for null values in each column
print(df.isnull().sum())
```

```
App                 0
Category            0
Rating           1474
Reviews             0
Size                0
Installs            0
Type                1
Price               0
Content Rating      1
Genres              0
Last Updated        0
Current Ver         8
Android Ver         3
dtype: int64
```

📌 STEP 4: Drop Null Values

In [16]:
```python
# Step 4: Drop records with nulls in any of the columns
df.dropna(inplace=True)
print("Null values after dropping:", df.isnull().sum().sum())

# Check again
df.isnull().sum()
```

```
Null values after dropping: 0
```

Out[16]:
```
App               0
Category          0
Rating            0
Reviews           0
Size              0
Installs          0
Type              0
Price             0
Content Rating    0
Genres            0
Last Updated      0
Current Ver       0
Android Ver       0
dtype: int64
```

📌 STEP 5: Data Cleaning

🔹 (A) Size Column Clean

In [17]:
```
# Step 5(A): Fixing Size column - Convert Kb to Mb and then to numeric
# Remove 'Varies with device'
df = df[df['Size'] != 'Varies with device']

# Convert Size to numeric
def convert_size(size):
    if 'M' in size:
        return float(size.replace('M','')) * 1000
    elif 'k' in size:
        return float(size.replace('k',''))
    else:
        return np.nan

df['Size'] = df['Size'].apply(convert_size)
df.dropna(subset=['Size'], inplace=True)
```

🔹 (B) Reviews → Numeric

In [18]:
```
# Step 5(B): Convert Reviews to numeric
df['Reviews'] = pd.to_numeric(df['Reviews'])
```

🔹 (C) Installs Clean

In [19]:
```
# Step 5(C): Fix Installs - Remove '+' and ',' and convert to int
df['Installs'] = df['Installs'].str.replace('+','')
df['Installs'] = df['Installs'].str.replace(',','')
df['Installs'] = df['Installs'].astype(int)
```

🔹 (D) Price Clean

In [20]:
```
# Step 5(D): Fix Price - Remove '$' and convert to numeric
df['Price'] = df['Price'].str.replace('$','')
df['Price'] = df['Price'].astype(float)
```

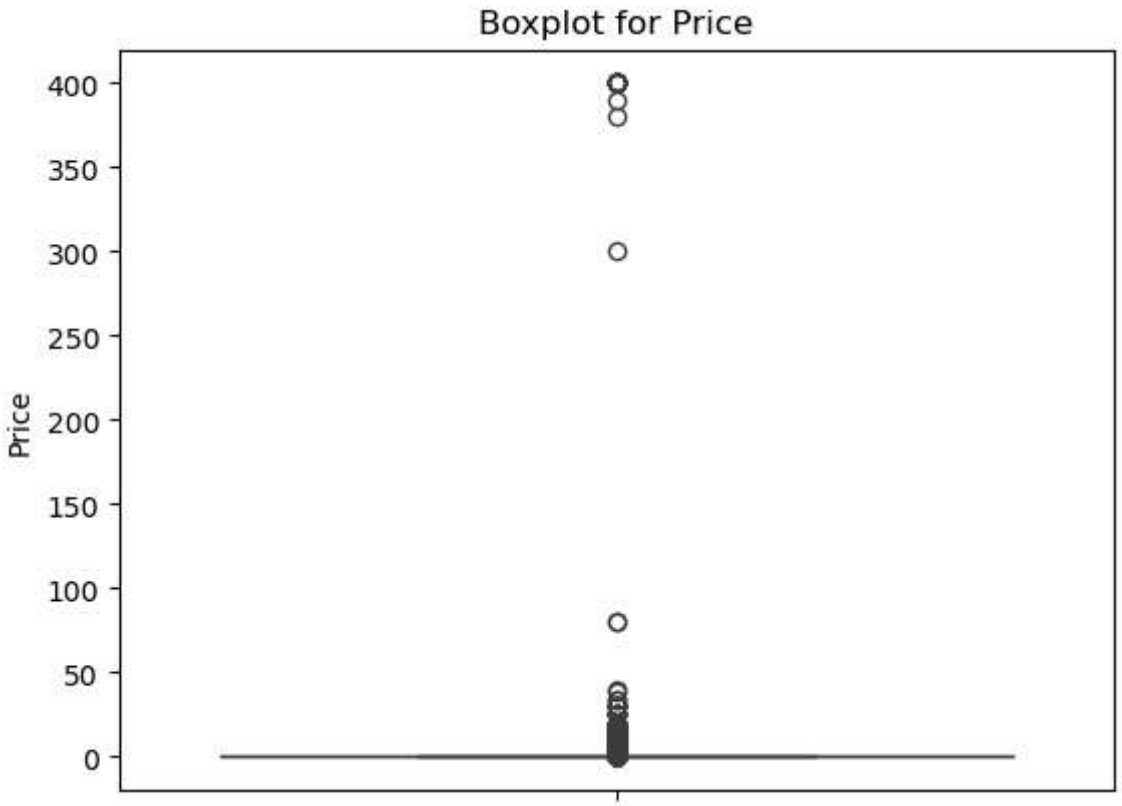
(E) Sanity Checks

In [21]:
```
# Step 5(E): Sanity Checks
# 1. Rating between 1 and 5
df = df[(df['Rating'] >= 1) & (df['Rating'] <= 5)]

# 2. Reviews should not be more than installs
df = df[df['Reviews'] <= df['Installs']]

# 3. For Free apps, price must be 0
df = df[~((df['Type'] == 'Free') & (df['Price'] > 0))]
```

📊 STEP 6: Univariate Analysis

Boxplot Price

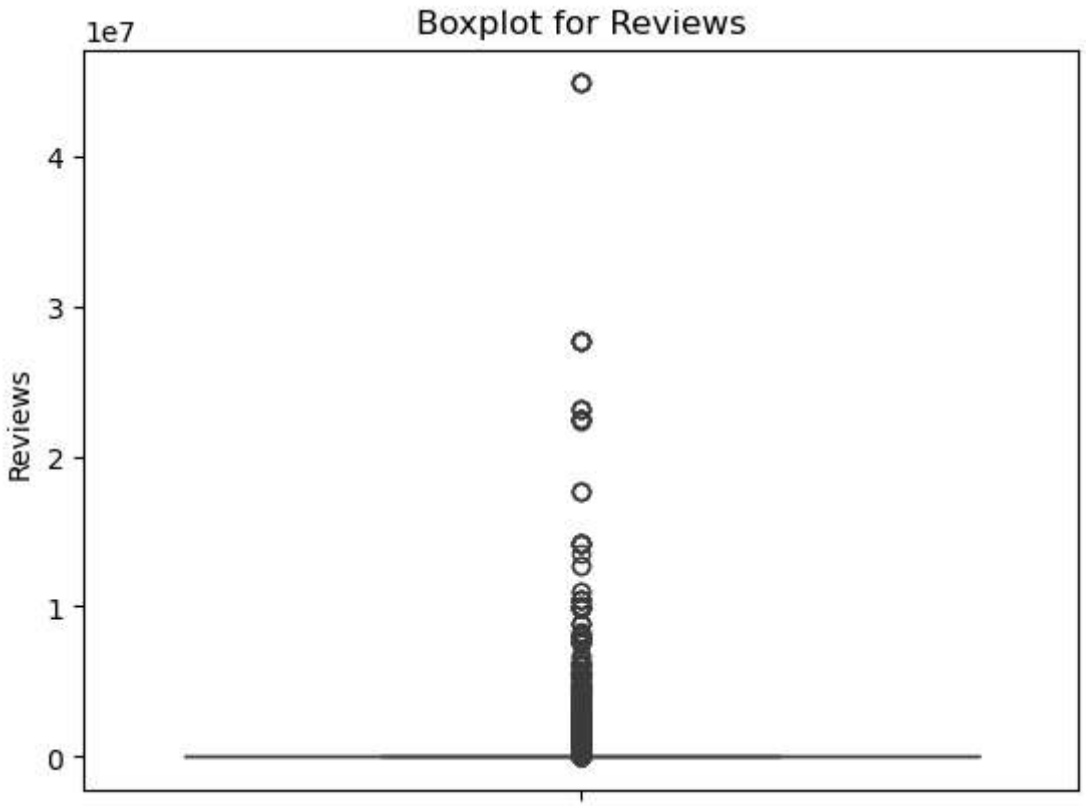
In [22]:
```
# Boxplot for Price
sns.boxplot(df['Price'])
plt.title("Boxplot for Price")
plt.show()
```




```
Observation: Most apps are free or low priced, few extreme outliers present
```

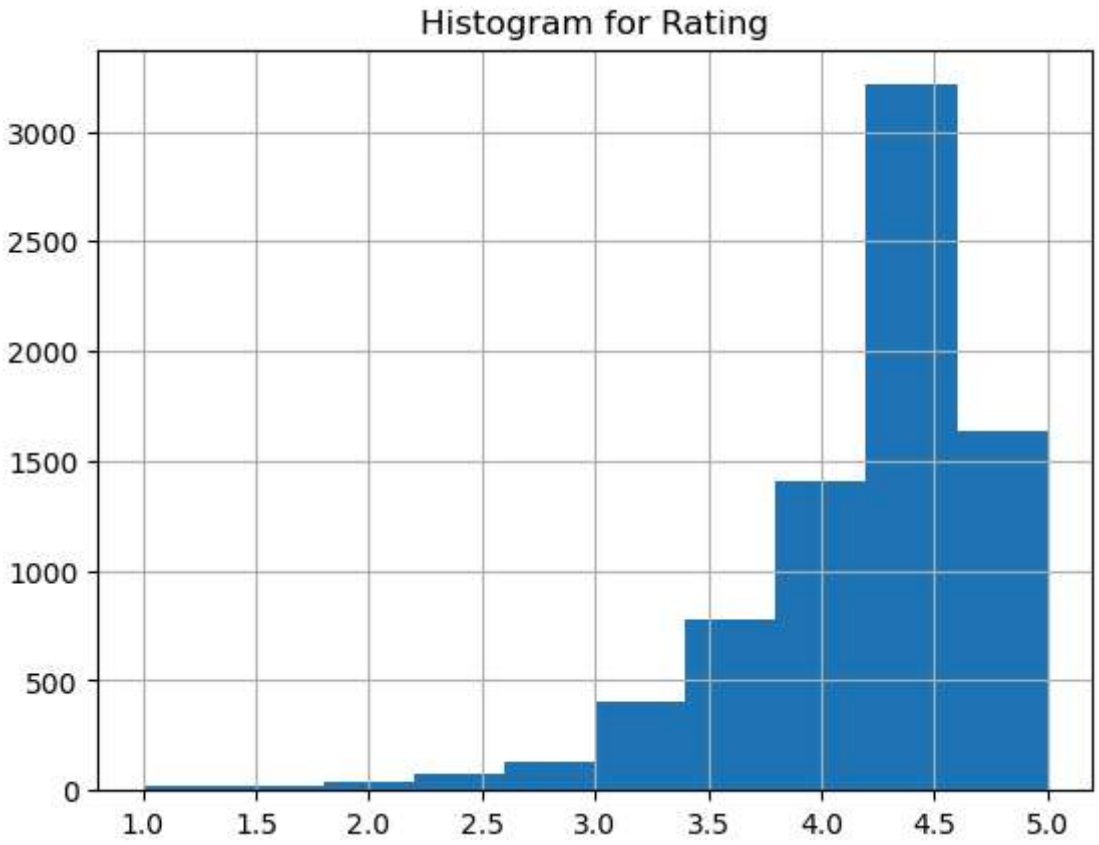
Boxplot Reviews

In [23]:
```
sns.boxplot(df['Reviews'])
plt.title("Boxplot for Reviews")
plt.show()
```



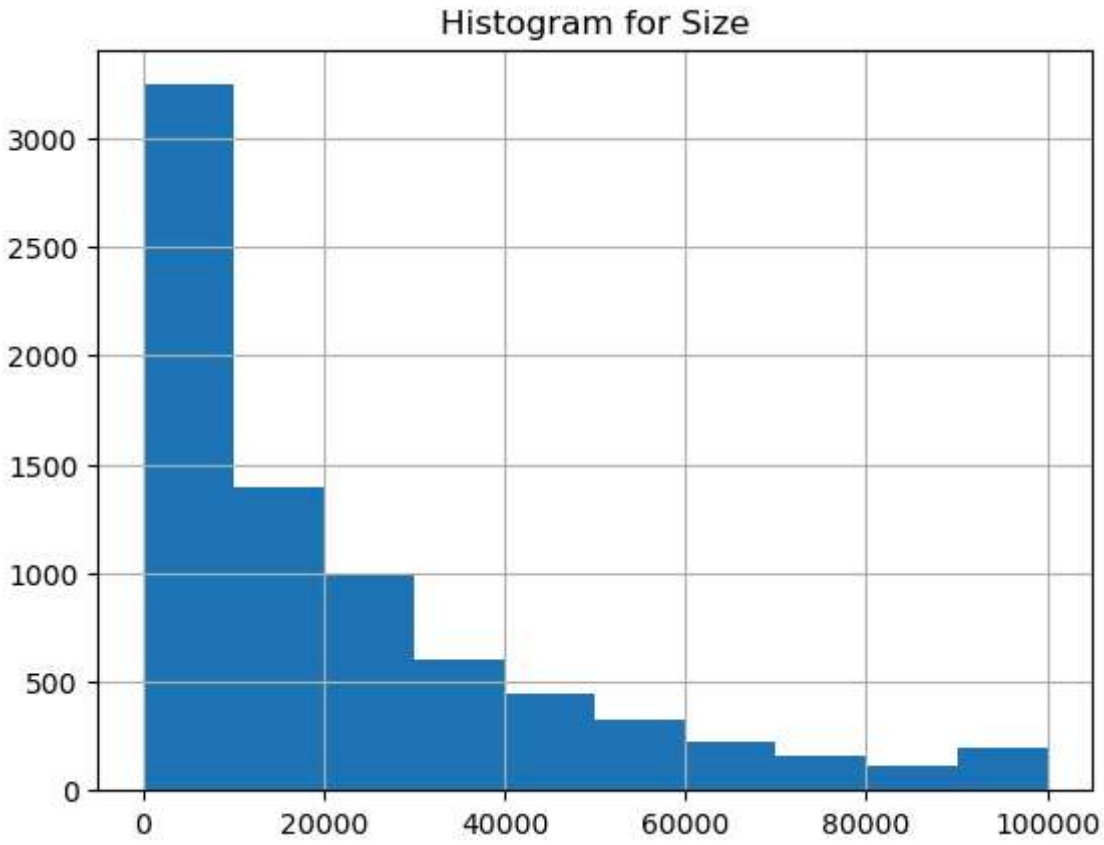
- ### Histogram Rating

In [24]:
```
df['Rating'].hist()
plt.title("Histogram for Rating")
plt.show()
```



Histogram Size

In [25]:
```
df['Size'].hist()
plt.title("Histogram for Size")
plt.show()
```



⚠️ STEP 7: Outlier Treatment

- **Price**

In [26]:
```
# Drop apps with price >= 200 (junk apps)
df = df[df['Price'] <= 200]
```

- **Reviews**

In [27]:
```
# Drop records with more than 2 million reviews
df = df[df['Reviews'] <= 2000000]
```

- ### Installs

In [28]:
```
# Drop high install apps based on percentile (cutoff)
# Finding 95th percentile for Installs
df['Installs'].quantile([0.1,0.25,0.5,0.7,0.9,0.95,0.99])
df = df[df['Installs'] <= df['Installs'].quantile(0.95)]
```

📊 STEP 8: Bivariate Analysis

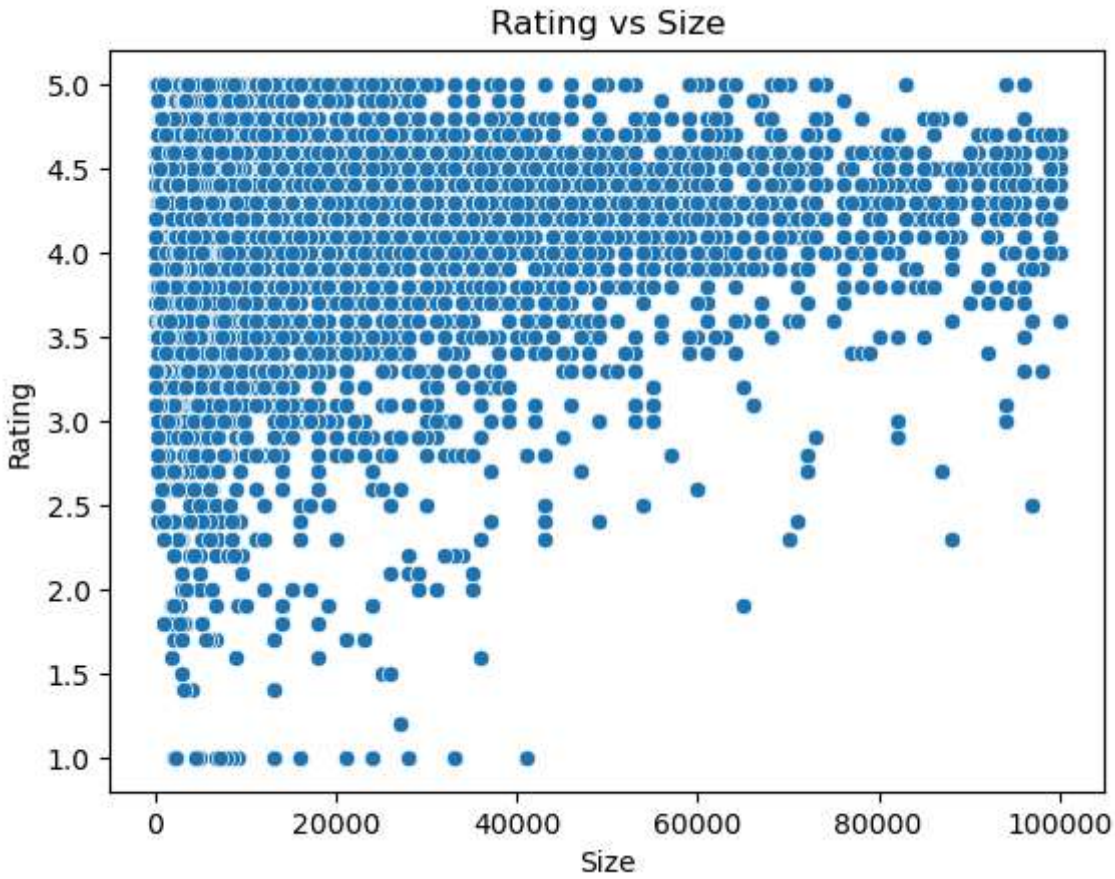
- ### Rating vs Price

In [32]:
```
sns.scatterplot(x='Price', y='Rating', data=df)
plt.title("Rating vs Price")
plt.show()
```

```
# Observation: No strong relationship between price and rating
```

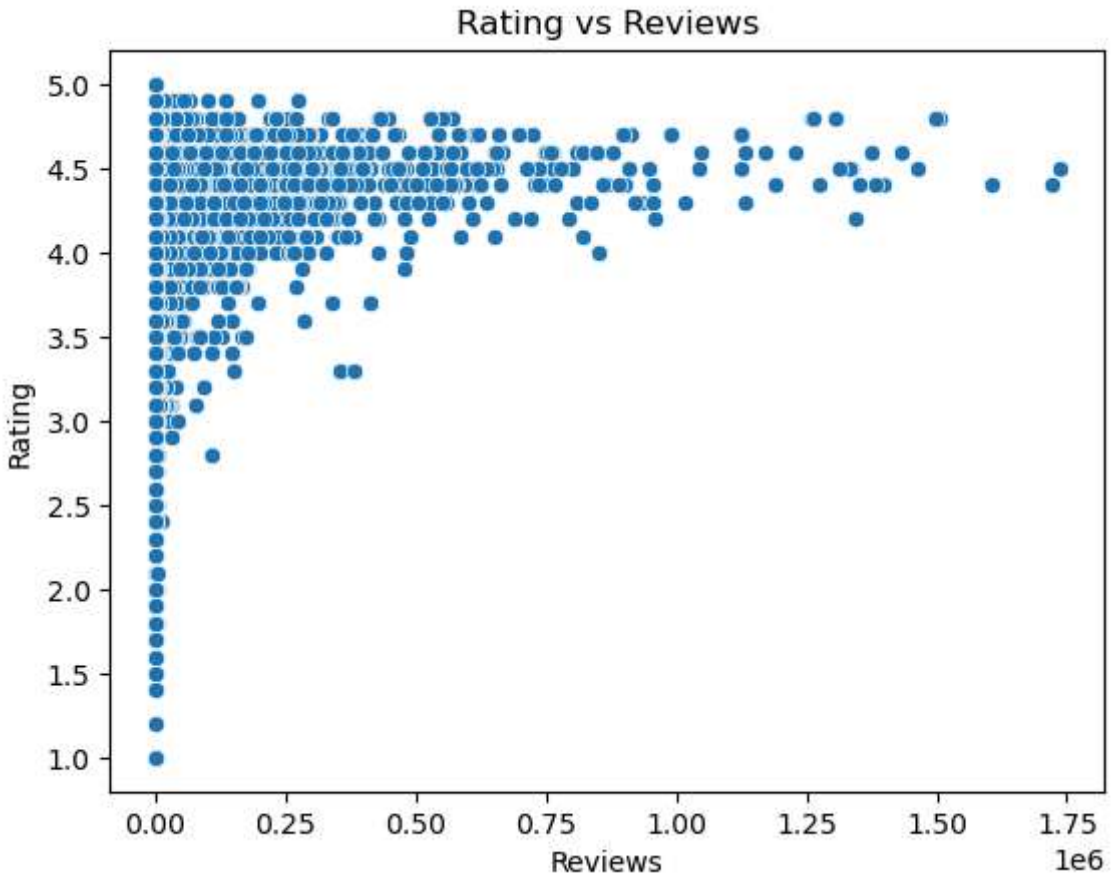
Rating vs Size

In [34]:
```
sns.scatterplot(x='Size', y='Rating', data=df)
plt.title("Rating vs Size")
plt.show()
```



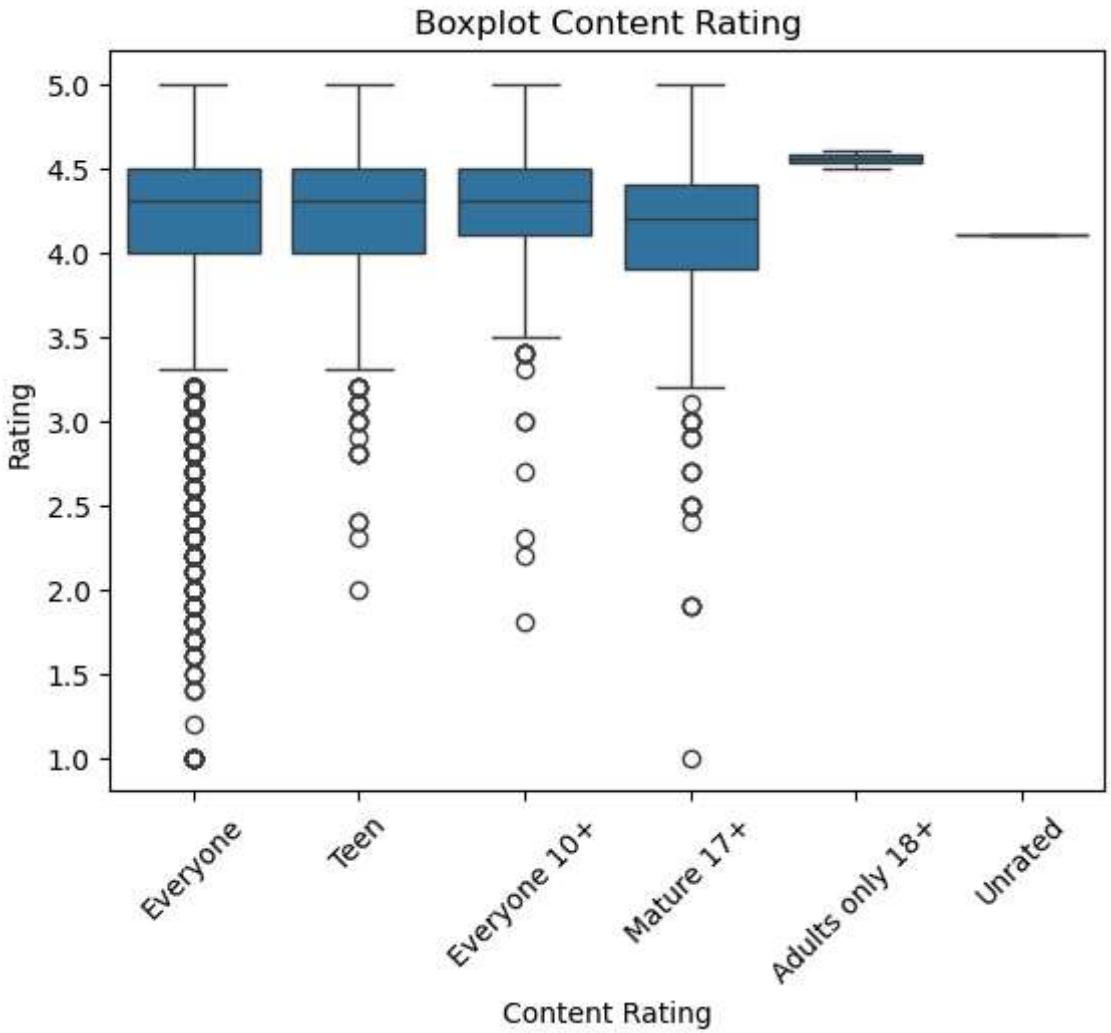
Rating vs Reviews

In [35]:
```
sns.scatterplot(x='Reviews', y='Rating', data=df)
plt.title("Rating vs Reviews")
plt.show()
```



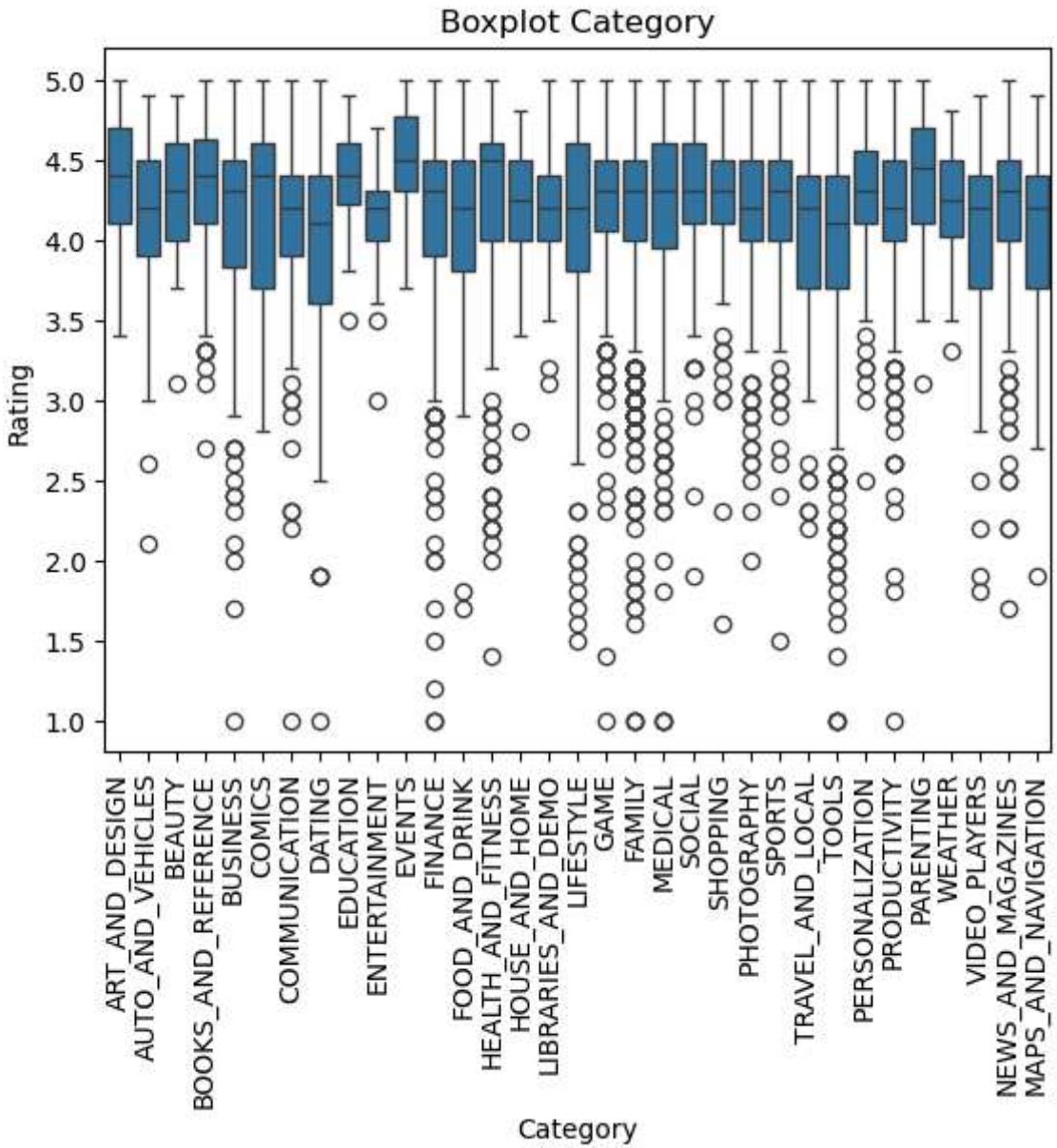
Boxplot Content Rating

In [36]:
```
sns.boxplot(x='Content Rating', y='Rating', data=df)
plt.xticks(rotation=45)
plt.title("Boxplot Content Rating")
plt.show()
```



Boxplot Category

In [37]:
```
sns.boxplot(x='Category', y='Rating', data=df)
plt.xticks(rotation=90)
plt.title("Boxplot Category")
plt.show()
```



STEP 9: Data Preprocessing

In [38]:
```
# Copy dataframe
inp1 = df.copy()
```

🔹 Log Transformation

In [39]:
```
# Applying log transformation to reduce skewness
inp1['Reviews'] = np.log1p(inp1['Reviews'])
inp1['Installs'] = np.log1p(inp1['Installs'])
```

🔹 Drop Columns

In [40]:
```
# Drop unnecessary columns
inp1.drop(['App', 'Last Updated', 'Current Ver', 'Android Ver'], axis=1, inplace=True)
```

🔹 Dummy Encoding

In [41]:
```
# Create dummy columns (One-Hot Encoding)
inp2 = pd.get_dummies(inp1, columns=['Category', 'Genres', 'Content Rating'], drop_first=True)
```

📊 STEP 10: Train Test Split

In [44]:
```
df_train, df_test = train_test_split(inp2, test_size=0.3, random_state=42)
```

📌 STEP 11: X and y

In [45]:
```
X_train = df_train.drop('Rating', axis=1)
y_train = df_train['Rating']

X_test = df_test.drop('Rating', axis=1)
y_test = df_test['Rating']
```

🤖 STEP 12: Model Building

In [57]:
```python
# Create pipeline with preprocessing and model
model = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ('regressor', LinearRegression())
])

#Model Building - Linear Regression
model.fit(X_train, y_train)

# R2 score on train set
train_pred = model.predict(X_train)
r2_train = r2_score(y_train, train_pred)

print("Train R2 Score:", r2_train)
```

```
Train R2 Score: 0.16036440979501387
```

📊 STEP 13: Test Prediction

In [58]:
```python
# Test prediction
test_pred = model.predict(X_test)

# Test R2 score
r2_test = r2_score(y_test, test_pred)

print("Test R2 Score:", r2_test)
```

```
Test R2 Score: 0.11349269132011874
```

Conclusion: This model helps predict app ratings based on various features. Further improvements can be done using advanced models.